

The anti-oxidation and anti-aging effects of *Ganoderma lucidum* in *Caenorhabditis elegans*

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ABSTRACT

As a traditional herbal medicine, the clinical efficacy of *Ganoderma lucidum* (*G. lucidum*, also known as Lingzhi in China) has been proved by clinical research and a large number of animal experiments. However, its pharmacological mechanism is not clear. Here, we used the *Caenorhabditis elegans* as an animal model to study the anti-oxidative stress and anti-aging effects of *G. lucidum* water extract. Our results showed that *G. lucidum* effectively promoted the nematodes to resist the oxidative stress of paraquat and heavy metal Cr⁶⁺, and significantly prolonged the lifespan of the nematodes. The underlining mechanisms were further investigated by focusing on the signaling pathways that regulate the stress responses and the lifespan. We found that *G. lucidum* protected the nematode against the insults of paraquat and heavy metals through the diet restriction pathway and the mTOR/S6K signaling pathway, respectively. Whereas, the effect of *G. lucidum* on the longevity of the nematode mainly depended on the germline signaling pathway. Microarray assays were conducted to reveal the gene expression profiles. The expression levels of 2746 genes were significantly changed during the aging process, of which 34 genes were reversed in their expression by the treatment of *G. lucidum* in aged nematodes. These results suggest that *G. lucidum* regulates the biophysiological processes in the nematodes through multiple signaling pathways.

1. Introduction

Ganoderma lucidum (Lingzhi in China) has been used for medicinal purposes by Chinese for more than thousand years. As a traditional herbal medicine (THM) in China, *G. lucidum* is still widely used nowadays as an adjuvant treatment for many diseases in clinic. This THM is considered to exert the exact clinical effects by regulating immunity, reducing inflammatory response, anti-aging and so on (Huang and Ning, 2010; Hasnat et al., 2015; Barbieri et al., 2017; Weng et al., 2010). Although the medical value of *G. lucidum* is well recognized, its pharmacological mechanism is largely unclear. Meanwhile, due to the lack of the corresponding relationship between active substances and biological effects, quality control of *G. lucidum* products is difficult. The bioactive substances and pharmacological studies of *G. lucidum* depend on an efficient biological model. *Caenorhabditis elegans* (*C. elegans*) is an important model organism which is commonly used in life science to study basic biological processes, such as aging, innate immunity, nervous system function (Kurz and Tan, 2004; E et al., 2018; Honjoh et al., 2017). Due to the conserved biological pathways between nematodes and mammals, the knowledge obtained in the nematode is mostly

applicable to mammals including mankind (Antikainen et al., 2017; Hengartner and Horvitz, 1994; Castel and Martienssen, 2013; Burkewitz et al., 2016). *C. elegans* as an *in vivo* model has also been successfully used to clarify the pharmacological mechanisms in drug discovery and screening (Kumar and Lombard, 2016; Cabreiro et al., 2013a).

In this paper, we examined the effects of *G. lucidum* water extract on extending lifespan and enhancing stress resistance in *C. elegans* and revealed the underling signal pathways using classic genetic analysis.

2. Materials and methods

2.1. Nematode strains

Caenorhabditis elegans Bristol strain N2, CB1370, CF1038, CF1903, TJ1052, RB1206, DA465 and *Escherichia coli* OP50 were provided by the *Caenorhabditis elegans* Genetics Center, University of Minnesota (Caenorhabditis Genetics Center/CGC). Bisexual strains were used for drug testing and genetic analysis. They were grown on solid (NGM) or liquid (S-Medium) nematode growth medium seeded with *E. coli* OP50

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(Brenner, 1974). The temperature for nematodes maintenance was set at 20 °C unless stated otherwise.

2.2. Water extract of *G. lucidum*

Thirty grams of dried *G. lucidum* (Tong Ren Tang Pharmacy, Shanghai) were ground and extracted in 420 mL of boiling dd H₂O for 2 h. The water extract was collected by centrifuge and concentrated to 100 mL by vacuum drying. The concentration of the water extract was defined as 300 mg/mL (substance in 1 mL was from 300 mg original dried *G. lucidum*). The water extract was aliquoted and stored in –20 °C until use.

2.3. Oxidative stress assays

Age-synchronized eggs were obtained by bleaching of adult worms with lysis solution. The eggs were hatched and grown up to stage of L4 on NGM plate. The L4 larvae were then transferred to S-Medium supplemented with OP50 and 400 µM FUDR in a 25 cm² cell culture flask. After 24 h, *G. lucidum* extract was added to a required concentration and maintained for 5 days. Then, the S-Medium was removed and the worms were rinsed three times with M9 buffer to wash out the residual *G. lucidum* extract and OP50. For the assay of oxidative stress, about ninety nematodes were selected and divided into three wells of 48-well culture plate containing M9 buffer with 100 mM paraquat or 10 mM K₂Cr₂O₇. The plate was shaken every 5 h and the fraction of nematodes alive was scored.

2.4. Lifespan assay

Lifespan assay was performed as previously reported (Petrascheck et al., 2007). Age-synchronized nematodes were transferred to a 96-well plate as L1 larvae together with 6 mg/mL OP50 in 120 µL S-Medium. Each well contained 8–14 worms, and about 60 worms were used in each group for an individual assay. The plate was kept in the incubator at 20 °C. 15 µL FUDR was added in 48 h to prevent self-fertilization (0.12 mM final), and 15 µL different concentration of *G. lucidum* extract solution was added 20 h later (day 1 of adulthood). The lifespan of each worm was recorded on the basis of body movement once every two days.

2.5. Nile Red staining

Worms were treated as in the lifespan assay. The treated worms were harvested in 1.5 mL eppendorf tube and washed by M9 buffer for three times. 50 µL of freshly prepared 10% paraformaldehyde solution was added. The liquid was treated with cyclic liquid nitrogen freezing–thawing for two times. Then the worms were centrifuged and supernatant was removed. The worms were stained with 5 µg/mL Nile Red for 30 min. Then the worms were washed with M9 buffer and transferred onto a 2% agarose pad for microscopic observation and photography (Brooks et al., 2009; Liu et al., 2014). Image-pro plus 6.0 was used to analyze the fluorescence intensity.

2.6. Microarray assay and bioinformatics analysis

The worms at day 10 and day 22 of adulthood with or without the treatment of *G. lucidum* extract were collected. The dead worms were picked out and the rest were stored frozen in liquid nitrogen. Each group was biologically duplicated. All 8 groups of samples were subjected to the whole transcriptome microarray assay (Shanghai Biotechnology Company, Shanghai, China). Gene expression profile was assessed by Affymetrix 3' IVT Expression Array. Limma package in R was used to determine the differentially expressed genes (DEGs) according to the normalized gene expression data. Genes with a fold change of 2 or more and FDR (false positive rate) of 0.05 or less were

regarded as significantly differentially expressed ones. Then Cluster and TreeView program were used to draw the heatmap to show the expression pattern of the DEGs.

2.7. Statistical analysis

Data were expressed as mean ± SEM. In lifespan assay, the significance of comparison was calculated by Log-rank (Mantel-Cox) Test, and others by Student's *t*-test unless mentioned otherwise. Statistical analysis was performed with Prism version 5.0 (GraphPad Software, CA, USA). Results were considered to be statistically significant at *P* value < 0.05. Each of the assays was repeated at least two independent times with similar results, and the data of representative experiments are shown.

3. Result

3.1. *G. lucidum* increases the ability of the nematode against oxidative stress

Paraquat and heavy metal Cr⁶⁺ were used to evaluate the responses to oxidative stress in *C. elegans*. The concentrations of *G. lucidum* extract used were 0, 0.75, 3.75, 7.5, 11.25 and 15 mg/mL. Pretreatment of *G. lucidum* extract markedly increased the ability of the nematode against paraquat and Cr⁶⁺ insults in the survival assays (Fig. 1A, C) and the median survival time (Fig. 1B, D). The effects of the extract on the suppression of paraquat- and Cr⁶⁺-induced oxidative stress showed a dose-dependent profile. The extract at a concentration of 7.5 mg/mL exhibited the peak effect, which increased median survival time of paraquat treated nematodes from 40.6 h to 71.3 h and maximum survival time from 85 h to 105 h. The same effect of the extract of 7.5 mg/mL was also observed on Cr⁶⁺ treated nematodes, the median survival time was extended from 26.7 h to 65 h and the maximum survival time was extended from 70 h to 90 h.

3.2. *G. lucidum* prolongs the nematode lifespan

Oxidative stress is involved in the process of aging. Further, the effect of *G. lucidum* extract on the nematode lifespan was tested. Pretreatment of *G. lucidum* extract significantly expanded the lifespan of worms, with a peak effect at 7.5 mg/mL (Fig. 2A, B). The effects of the extract on the lifespan extension showed a dose-dependent profile as well. Compared with the control group, *G. lucidum* extract at a concentration of 7.5 mg/mL increased the median lifespan from 17.3 days to 23.5 days and maximum longevity from 34 days to 38 days. However, higher concentrations of *G. lucidum* extract showed no, even deleterious effects (Fig. 2A, B).

The extract at a concentration of 7.5 mg/mL exhibited the peak effects on stress resistance and lifespan extension. Therefore, the concentration of 7.5 mg/mL was chosen for the following experiments.

By Nile Red staining, we found the extract markedly reduced the body fat content (Fig. 2C). Furthermore, the extract treatment also reduced the number of eggs laid (Fig. 2D), but significantly extended the laying period (Fig. 2E).

3.3. Signaling pathways essential for the anti-oxidative effect of *G. lucidum*

In order to reveal the signal pathways involved in the protective effects of *G. lucidum*, we adopted the classic genetic analysis strategy and focused on *daf-16*, *daf-2*, *glp-1*, *eat-2* and *rsk-1* gene mutations. DAF-2 and its downstream transcription factor DAF-16/FOXO are the important components of insulin signal pathway which regulates a large set of genes related to stress (Oh et al., 2006). *G. lucidum* extract dramatically improved the survival of both *daf-16* and *daf-2* mutants under the stress of paraquat-induced oxidation (Fig. 3A, B). Similarly, the protective effect maintained in *glp-1* (Fig. 3C) and *rsk-1* (Fig. 3E) mutant worms. However, the increasing survival effect of the extract in

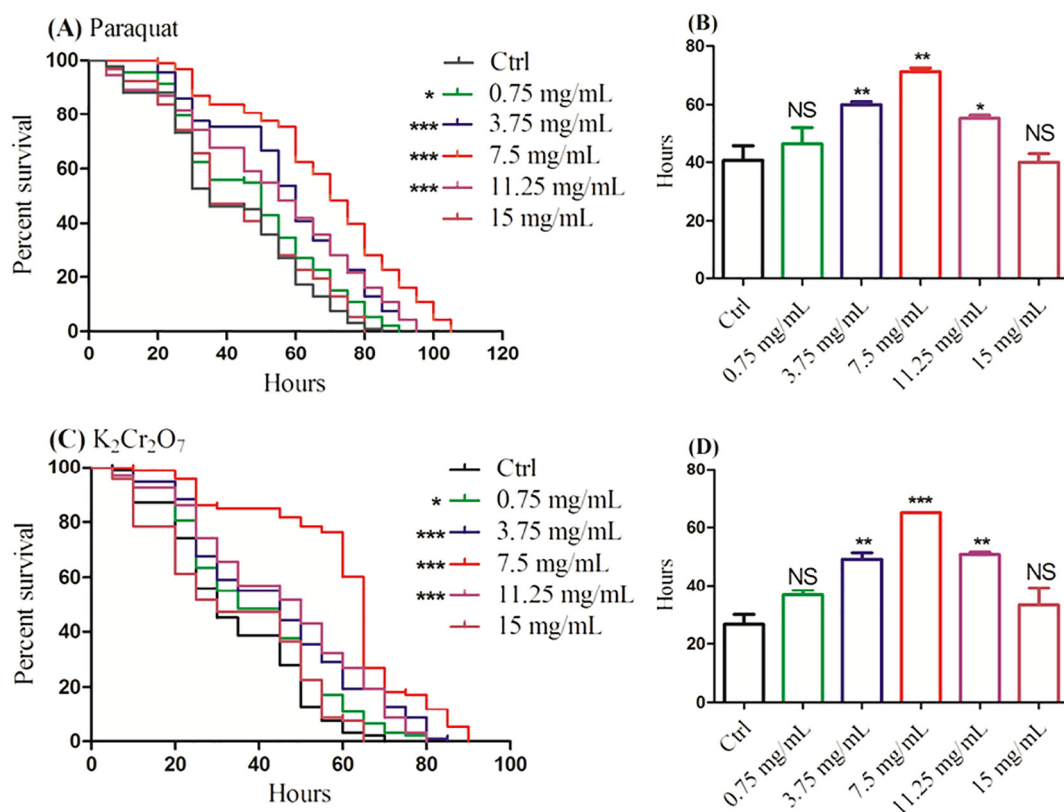


Fig. 1. The anti-oxidative stress effect of *G. lucidum* in the nematode.

Pretreatment of *G. lucidum* extract protected the wild type N2 nematode against paraquat (A, B) and heavy metal Cr⁶⁺ (C, D). (A, C) The survival curves; (B, D) The median survival time. **P* < 0.05; ***P* < 0.01; ****P* < 0.001.

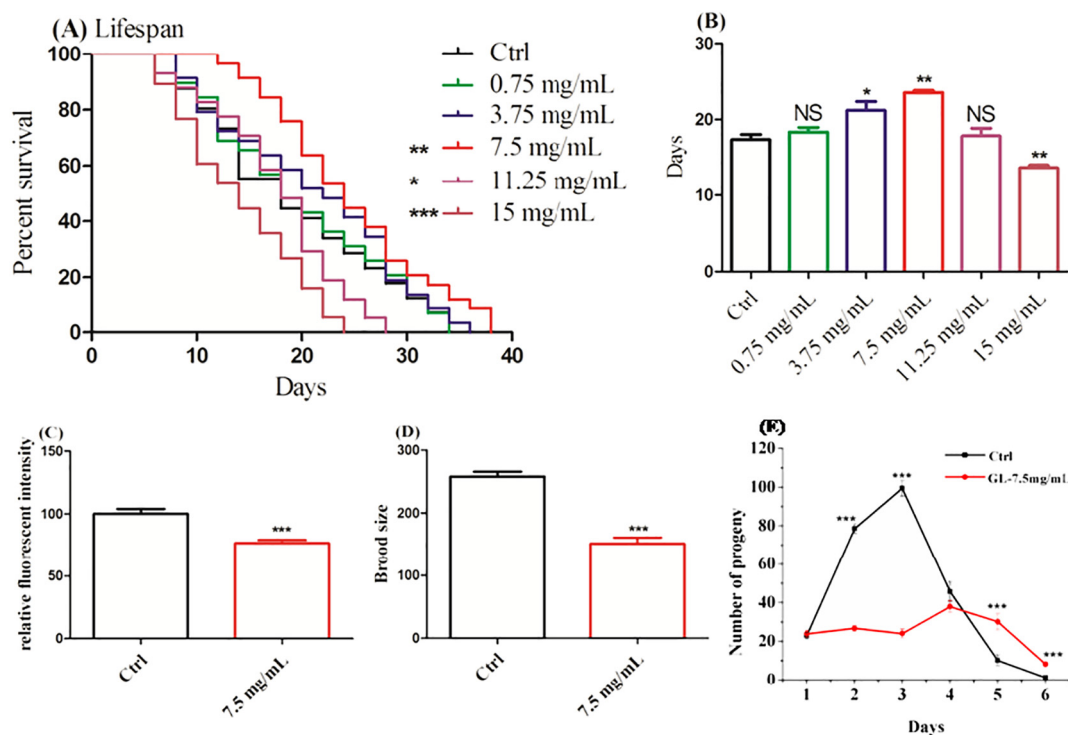


Fig. 2. The anti-aging effect of *G. lucidum* in the nematode.

Treatment of *G. lucidum* extract prolonged the lifespan of wild type N2 nematode (A, B). (A) The survival curve. (B) The median survival days. (C) Treatment of *G. lucidum* extract reduced the body fat content. (D) Treatment of *G. lucidum* extract reduced the number of eggs laid. (E) Treatment of *G. lucidum* extract extended the laying period. **P* < 0.05; ***P* < 0.01; ****P* < 0.001.

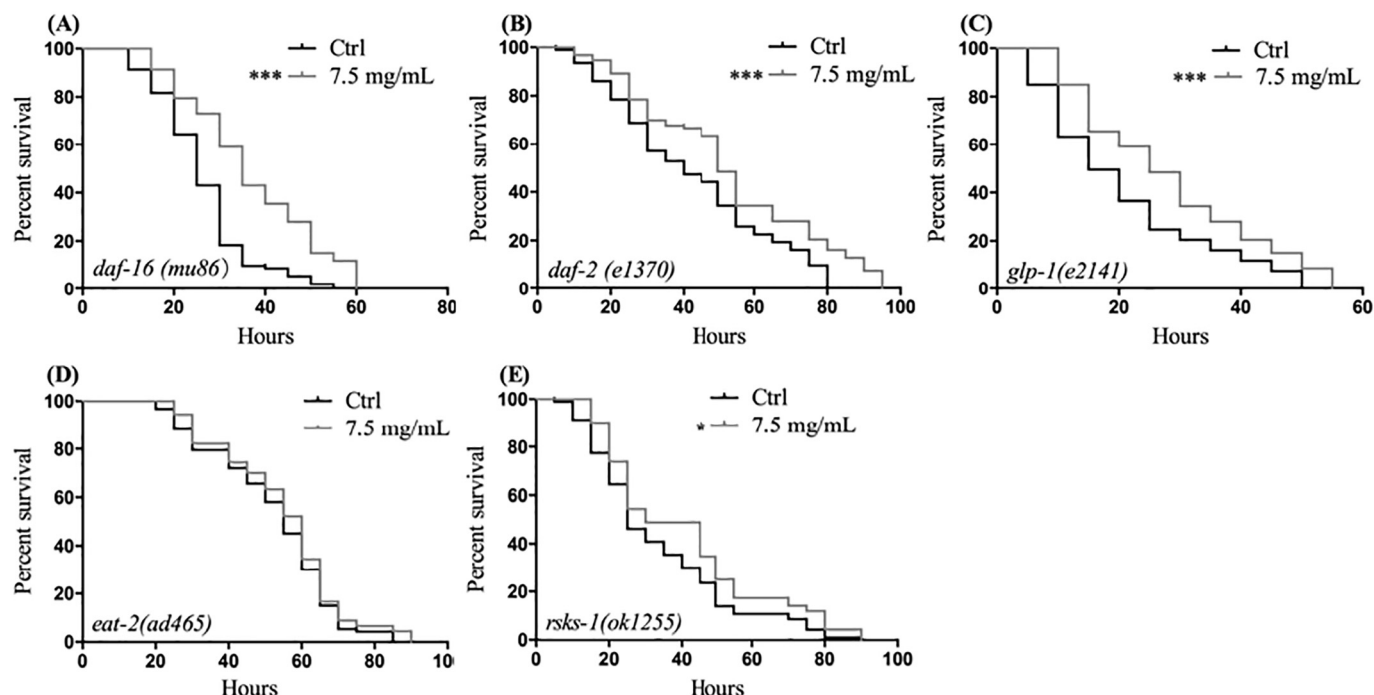


Fig. 3. The effects of *G. lucidum* extract in gene mutation strains challenged with paraquat.

(A) *daf-16(mu86)*. (B) *daf-2(e1370)*. (C) *glp-1(e2141)*. (D) *eat-2(ad465)*. (E) *rsk-1(ok1255)*. * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$.

paraquat-induced oxidative stress model was vanished in *eat-2* mutant worms, suggesting this effect is dependent on diet restriction signaling pathway (Fig. 3D).

In model of Cr^{6+} induced-oxidative stress, *G. lucidum* extract markedly improved the survival of *daf-16*, *daf-2*, *glp-1* and *eat-2* mutants (Fig. 4A, B, C, D), whereas the effect was abolished in *rsk-1* mutant worms (Fig. 4E), which suggests that mTOR/S6K signaling pathway is indispensable for the survival promoting effect against the stress of heavy metal.

3.4. Signaling pathway involved in the anti-aging effect of *G. lucidum*

The lifespan of the nematode is regulated by multiple signaling pathways. In this study, the contribution of signaling pathways including insulin, germline, dietary restriction, and mTOR/S6K pathways to the anti-aging effect of *G. lucidum* were investigated. Lifespan extension was observed in *daf-16*, *daf-2*, *eat-2* and *rsk-1* mutant worms, but not in *glp-1* mutants (Fig. 5). Therefore, the effect of *G. lucidum* extract on longevity depended on the germline signaling pathway.

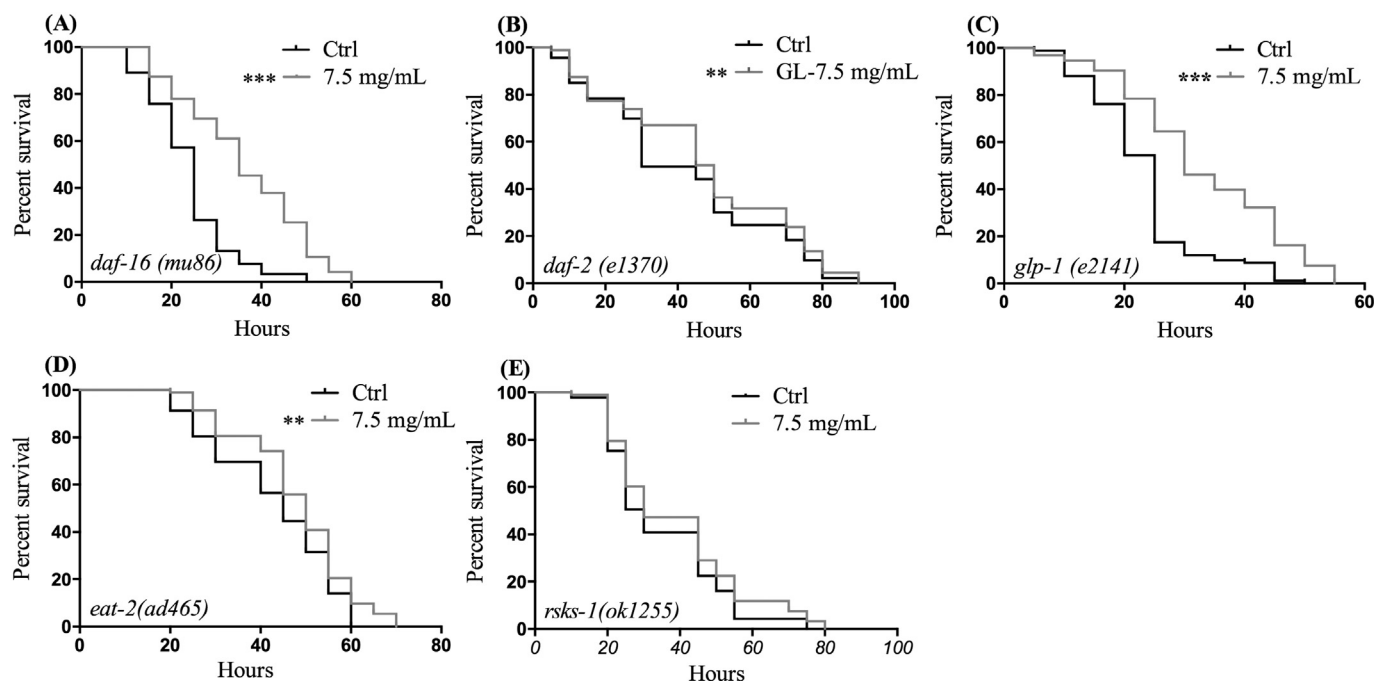


Fig. 4. The effects of *G. lucidum* extract in gene mutation strains challenged with heavy metal Cr^{6+} .

(A) *daf-16(mu86)*. (B) *daf-2(e1370)*. (C) *glp-1(e2141)*. (D) *eat-2(ad465)*. (E) *rsk-1(ok1255)*. * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$.

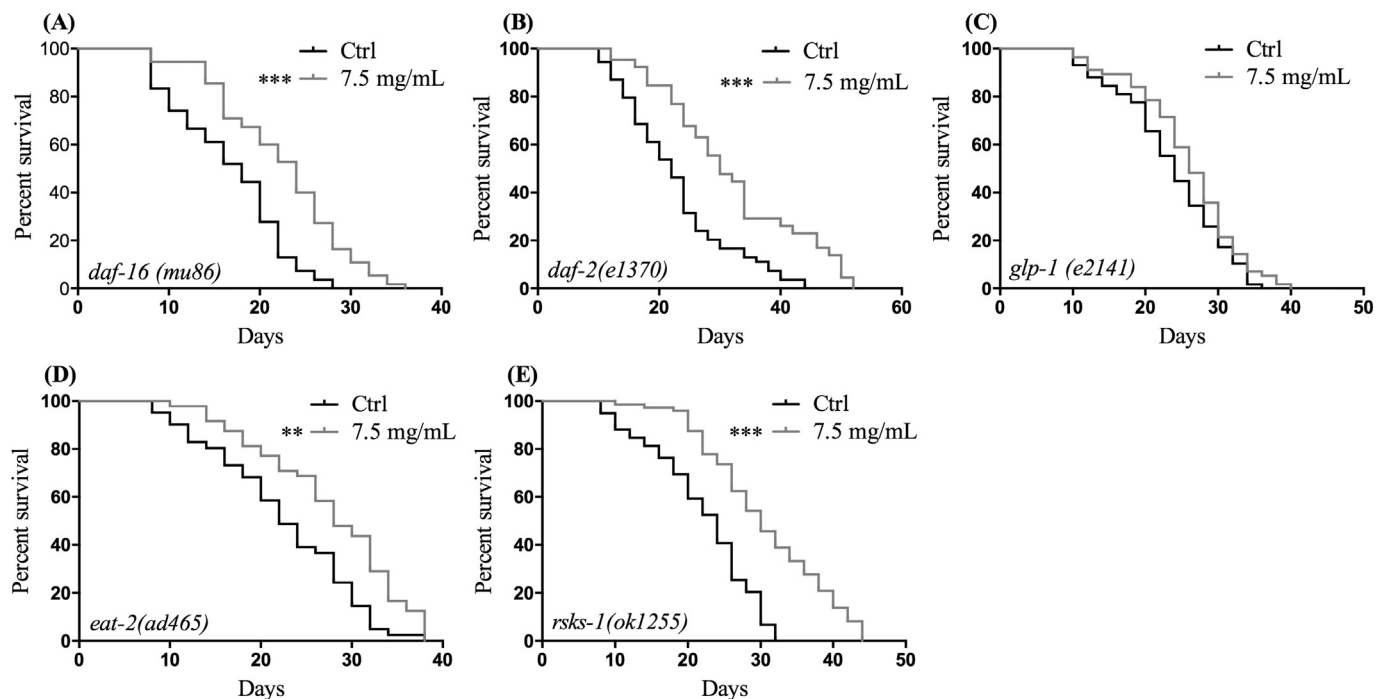


Fig. 5. The effects of *G. lucidum* extract on the lifespan of gene mutation strains.

(A) *daf-16(mu86)*. (B) *daf-2(e1370)*. (C) *glp-1(e2141)*. (D) *eat-2(ad465)*. (E) *rsk-1(ok1255)*. * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$.

3.5. *G. lucidum* rejuvenates the gene expression profiles in elder *C. elegans*

Further, the effect of *G. lucidum* on the whole genome expression profile was analyzed by microarray assays. 2746 genes with expression level changed over 2 times during aging process from 8-day-old to 22-day-old, were identified (2082 genes down-regulated and 664 up-regulated) (Fig. 6A). Of these, expression pattern of 34 genes in aging process was reversed in nematode by *G. lucidum* extract (Fig. 6B). Among them, 25 genes that were down-regulated during the aging were significantly increased after the treatment with *G. lucidum* extract and nine genes up-regulated during the aging were inhibited after the extract administration. More than half genes are not well studied or without any known functions. Only the *smg-1*, a serine-threonine kinase which plays a conserved role in nonsense-mediated mRNA decay (NMD) in worms and mammals, has been reported involved in longevity of *C. elegans* (Masse et al., 2008).

4. Discussion

G. lucidum, as a herbal medicine for longevity, has been used for thousands of years in China and East Asian countries. Recent studies have shown that *G. lucidum* displays auxiliary therapeutic effects on many diseases, including diabetes, cancer, inflammation, nervous system disease and many others. However, the mechanism of its pharmacological action is not clear. In this work, we showed that water extract of *G. lucidum* could effectively improve the resistance of nematodes to paraquat and heavy metal Cr^{6+} induced oxidative stress, and significantly prolong the life span of wild type nematodes. Genetic studies indicated that the effects of *G. lucidum* in nematodes might be contributed to different signaling pathways. The germline signaling played an essential role in the effect of *G. lucidum* on extending the lifespan of nematodes. Meanwhile, the treatment of *G. lucidum* in nematodes significantly reduced their eggs laid and extended the spawning period, further indicating the effect of *G. lucidum* on reproductive system. The role of *G. lucidum* in promoting nematode resistance to heavy metal injury was mainly dependent on the mTOR/S6K signaling pathway and the effect of anti-oxidative stress induced by

paraquat depended on EAT-2 dietary restriction pathway. These results suggest that *G. lucidum* can regulate the biophysiological processes in nematodes through multiple signaling pathways. A polysaccharide component in *G. lucidum* has been reported to prolong the lifespan of the nematodes by increasing Daf-16 expression through TIR-1 receptor and MAPK pathway (Chuang et al., 2009). In our experiment, the mechanism of *G. lucidum* extending the lifespan of nematodes does not depend on DAF-16. The possible reason is that we used the total water extract of *G. lucidum*, which contained components other than polysaccharides and polysaccharides *per se* are a mixture. Therefore, the comprehensive mechanism of these substances could be different from that reported by Chuang et al.. However, it is worth noting that *glp-1* mutation causes the longevity of nematodes depending on the function of DAF-16 (Alper et al., 2010). In our research, the effect of *G. lucidum* on lifespan is dependent on GLP-1, but not entirely on DAF-16. This suggests that the germline signaling can also regulate the nematode lifespan through other pathways. That the germline signaling enhances the innate immunity of nematode has been shown to be independent of DAF-16 (Alper et al., 2010). In the cases of oxidative stress and heat stress, the germline signaling does not rely on DAF-16 as well (Libina et al., 2003; Yamawaki et al., 2008). Our previous study showed that although *G. lucidum* from different production region all promoted the resistance of nematodes to paraquat, the mechanism underlying could be different regarding the *daf-16* signaling pathway (Vu et al., 2016). *G. lucidum* used in this study was produced in Hainan, a southern province of China, and their role in increasing nematode resistance to paraquat is independent of *daf-16*. Therefore, the components in *G. lucidum* could vary due to the different production regions, and even different batches. The biological activity of *G. lucidum* measured on *C. elegans* could be a useful index to evaluate the quality of the product. The metabolic status of the *E. coli* OP50, the food for *C. elegans*, plays a role on the growth and longevity of the nematode (Cabreiro et al., 2013b). Whether *G. lucidum* regulates *E. coli* metabolism and affects *C. elegans* growth is expected to be further studied.

Collectively, *G. lucidum* extract can regulate at least the dietary restriction, mTOR/S6K and germline signal pathways in *C. elegans*. Our work provides an important basis for further studies of the effective

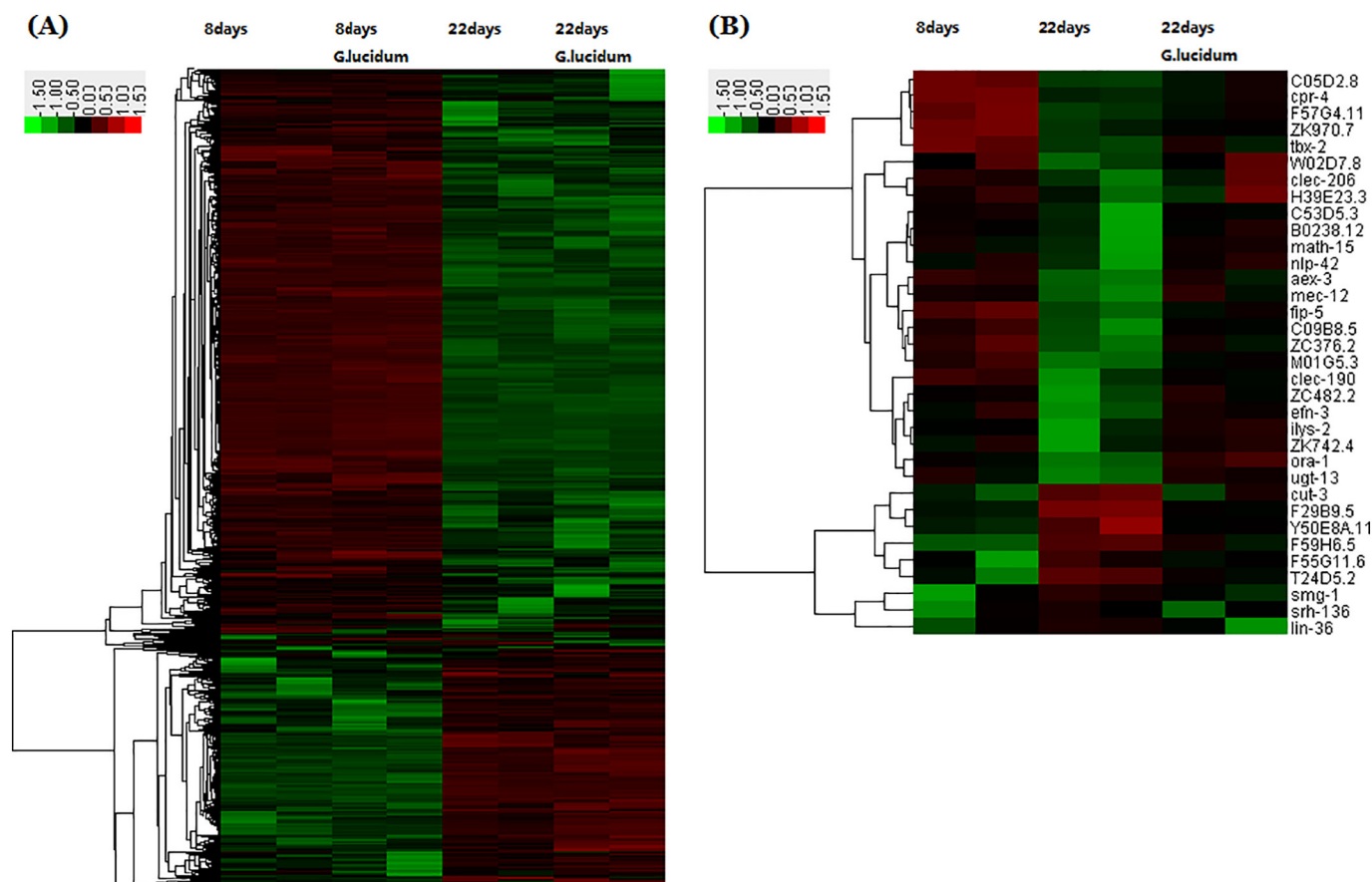


Fig. 6. Nematodes at 8-day-old and 22-day-old with or without *G. lucidum* treatment were subjected to whole genome expression profile assay. (A) Heat map of gene expression. (B) 34 genes reversed by *G. lucidum* treatment in aging process.

substance of *G. lucidum*.

Abbreviations used acknowledgment

G. lucidum (*Ganoderma lucidum*); *C. elegans* (*Caenorhabditis elegans*)

Qualification for authorship

VU THI CUONG, Ping Yang, Jian Fei designed the experiments; VU THI CUONG, Ping Yang, Weidong Chen performed the experiments; VU THI CUONG, Jiahao Shi, Mengjie Zhang, Hua Yang analyzed the data, Ning Wang, Sai Yang and Jianfeng Li gave technical helps.

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